

Proactive MLOps Control Addressing Entangled Enhancement in Machine Learning Pipelines

Abstract

Machine Learning Systems (MLS) frequently encounter “entangled enhancement,” a detrimental scenario where improving an individual model component paradoxically degrades overall system performance. This phenomenon underscores a significant limitation in current maintenance approaches that often rely on static policies, such as those analyzed by Wang and Machida [4] for two-component systems. Their work, while foundational, highlights the challenge of scaling such analytical methods to more complex, real-world MLS due to exponential growth in state-space complexity. This research plan details the development of an advanced MLOps framework designed to transcend these static policy limitations and offer a more scalable solution. The proposed framework will incorporate a predictive risk assessment module, trained on simulated data, to proactively identify the likelihood of entangled enhancement in multi-component systems. Based on these predictions, an adaptive retraining policy, potentially embodied by an intelligent agent or multi-agent system, will be employed, allowing for dynamic and context-aware decisions regarding model updates. This data-driven, adaptive approach is anticipated to offer a more computationally tractable and effective solution for managing complex interdependencies compared to direct analytical modeling for larger systems, thereby reducing the incidence of self-defeating improvements and significantly enhancing the stability, reliability, and operational efficiency of deployed AI systems.

Background and Motivation

Modern Machine Learning Systems (MLS) consist of interconnected models that must be frequently retrained due to dataset shifts—changes in real-world data that degrade performance. However, as shown by Wang and Machida [4], retraining does not always yield improvement. A key challenge is *entangled enhancement*—where improving an upstream model can unintentionally degrade overall system performance due to complex interdependencies.

This issue is not theoretical; large organizations like Meta report similar challenges in managing long dependency chains in production AI systems [3]. Wang and Machida analyzed two fixed maintenance strategies—*progressive* and *conservative*—in a two-component system using Continuous-Time Markov Chains (CTMCs). Their findings highlight that no single strategy is universally optimal; effectiveness varies with system parameters.

This proposal addresses the following critical gaps:

1. **Scalability Limitations:** CTMC-based analysis does not scale well to real-world MLS with many components due to state-space explosion. A more scalable approach is needed.
2. **Lack of Predictive Mechanisms:** Prior work focused on evaluating static policies post-hoc, without tools to predict entangled enhancement risks before retraining.
3. **Need for Adaptive Strategies:** Static policies are insufficient for dynamic environments. MLS maintenance must become adaptive, guided by real-time state and risk prediction.

This research builds on Wang and Machida’s work by proposing an MLOps framework that is scalable, predictive, and dynamically adaptive—better suited to the complexities of real-world AI systems.

Problem Statement

The central challenge this research confronts is the phenomenon of “entangled enhancement” within Machine Learning Systems composed of multiple, interconnected models. This is the often frustrating outcome where an upstream model, after being retrained and demonstrating improvement on its own specific metrics, unintentionally causes performance to decline or stagnate in downstream models or in the overall system’s output [4]. This occurs because downstream models have, in effect, learned to operate based on the specific output characteristics (including any idiosyncrasies) of the previous

version of the upstream model. Such “self-defeating improvements” result in significant practical negative consequences: valuable computational resources and skilled engineering time are expended on retraining efforts that do not yield the anticipated system-level benefits [1]; the reliability of AI-driven products can be compromised when seemingly positive updates lead to negative outcomes; and diagnosing system-wide performance issues becomes exceedingly difficult, particularly when localized “improvements” obscure these deeper, interaction-based failures.

The fundamental deficiency in many current MLOps practices [2, 4] is the absence of an intelligent, forward-looking mechanism specifically designed to anticipate and mitigate the risk of entangled enhancement before a decision is made to retrain an upstream model. This is especially true for systems that are more complex than the two-component model for which static policies were analyzed by Wang and Machida.

Research Objectives

The mission is to develop a practical and intelligent MLOps solution that directly addresses the challenge of entangled enhancement, moving beyond the limitations of static policy analysis. Specifically, the objectives are:

1. To develop a predictive model that estimates the risk of entangled enhancement resulting from retraining a specific upstream model in a multi-component Machine Learning System.
2. To design and implement an adaptive retraining policy that uses real-time entanglement risk predictions to guide retraining decisions within MLOps workflows.

Hypothesis

The following hypotheses will guide this investigation:

1. Primary Hypothesis (The main focus of this work): An MLOps system that integrates a predictive model for entangled enhancement risk and employs an adaptive retraining policy, driven by an intelligent agent framework and capable of actions such as deferring retraining, coordinating updates, or constraining changes based on this risk, will lead to a statistically significant reduction in system performance problems due to self-defeating improvements. It is also hypothesized that this approach will achieve higher overall system availability compared to static retraining policies in complex MLS environments.
2. Scalable Prediction Hypothesis: It is feasible to develop a predictive model that, by learning from features describing inter-component dependencies and historical patterns in a simulated complex MLS, can accurately forecast entangled enhancement likelihood. This data-driven predictive approach is anticipated to offer a significantly more scalable method for risk assessment in larger systems compared to analytical CTMC-based evaluations, which face exponential state-space complexity (such as 2^N states for N components as in [4]), reducing analytical complexity for larger systems.
3. Adaptive Agent Effectiveness Hypothesis: An intelligent agent (or multi-agent system) embodying the adaptive retraining policy, dynamically selecting interventions based on predicted entanglement risk, will demonstrate superior system stability and resource efficiency across diverse operating conditions compared to any single, fixed retraining policy.

Proposed Methodology

The research plan is centered on rigorously testing the Primary Hypothesis. The conviction is that by enabling MLOps systems to anticipate and intelligently respond to the specific risk of entangled enhancement, AI systems can be made more dependable and resource-efficient. This work aims to translate the foundational insights from Wang & Machida [4] into a practical, scalable MLOps solution. The plan unfolds as follows:

Phase 1: Building the “Entanglement Forecaster” - The Predictive Risk Engine

Before the MLOps system can act intelligently, it needs the ability to foresee potential trouble. This phase is about creating that predictive capability.

- **Crafting a Dynamic MLS Playground (Simulation Environment & Data Generation):**
Technology Choices & Feasibility: This simulated AI environment will be developed using Python. To manage the intricate interactions and state transitions in an MLS with multiple components (thereby extending beyond the direct applicability of CTMC analysis for two-component systems as in [4]), a discrete-event simulation library such as SimPy is planned. Should SimPy not fully meet the needs, a custom event-driven framework will be developed. This approach is highly feasible. This setup will allow for the creation of diverse MLS architectures, control over how data characteristics change over time, definition of various types of model interdependencies, and simulation of the conditions that give rise to entangled enhancement. To enhance fidelity, these interdependencies will be modeled not only as direct data flows but also by simulating nuanced relationships such as shared operational contexts or the influence of common upstream data sources on multiple components. The simulator will also parameterize the sensitivity of downstream models to subtle shifts in upstream output distributions, reflecting real-world interaction complexities. The controlled nature of this simulation makes the generation of sufficient, varied synthetic data for training the predictive model a highly feasible task.
The Process: This simulator will serve as the data generation engine, producing rich MLOps metadata. This will include records of retraining attempts, features of the involved models, metrics on data drift, and the observed outcomes of these updates - successful improvements, clear instances of entangled enhancement, or even cases of “degenerative retraining” where performance worsened.[4]
- **Identifying the “Red Flags” (Feature Engineering for Entanglement Prediction):**
Investigation will focus on features such as: the anticipated degree of change in an upstream model’s output distribution after retraining; historical data on how sensitive its downstream “neighbors” have been to its output variations; the complexity of the “data contract” or interface between connected models; and patterns observed in past retraining events involving these specific components. The feature engineering process will place particular emphasis on interaction terms and relational features. These are designed to specifically capture how changes in an upstream component’s output interface or statistical profile might mismatch the learned expectations of its direct downstream consumers, a characteristic uniquely indicative of entanglement risk beyond general component underperformance. This is a standard machine learning practice, and its feasibility depends on careful, iterative exploration.
- **Training the “Entanglement Forecaster” (Predictive Model Training & Complexity):**
Technology Choices & Feasibility: The primary toolkit will be Python, utilizing Scikit-learn for models like Gradient Boosting Machines or Random Forests. If relationships prove exceptionally complex, shallow Neural Networks via TensorFlow/Keras or PyTorch might be explored. Data manipulation will be handled using libraries such as Pandas and NumPy. Training these models is highly feasible.
Considering Computational Cost (Algorithmic Complexity & Feasibility): For a model like a Gradient Boosting Machine, the training time is generally manageable. Prediction is significantly faster, essential for timely MLOps decisions. This data-driven predictive approach offers a significant potential for complexity reduction in risk assessment for larger systems compared to analytical methods like CTMCs, which face exponential state-space complexity (such as 2^N states for N components as in [4]), making them computationally prohibitive for large, complex systems. The proposed predictive method aims to bypass this exponential complexity in the assessment phase.

Phase 2: Designing the “Smart Navigator” - The Adaptive Retraining Orchestrator (Agent-Based)

Knowing the risk is vital. The next step is to empower the MLOps system to use that knowledge for smarter decisions, which can be effectively realized through an intelligent agent framework.

- **Plugging into the MLOps Pipeline (Workflow Integration & Tech Stack):** The “Entanglement Forecaster” and the “Smart Navigator” agent system will be integrated into a representative MLOps pipeline.

Orchestration & Feasibility: Established open-source tools such as Apache Airflow or Kubeflow Pipelines are planned for workflow management. **Containerization & Feasibility:** Components will be packaged using Docker for consistency.

Tracking & Versioning & Feasibility: MLflow will be employed for experiment tracking, model versioning, and metadata logging. [2]

- **Defining Intelligent Actions (Policy Logic for the Agent System - Adaptive Actions):** When the system considers retraining an upstream model, the “Entanglement Forecaster” provides a risk score. The “Smart Navigator” intelligent agent (or a coordinating multi-agent system) will then evaluate this score and the broader system context to choose an action from its policy playbook. This involves: Low Risk: The agent directs the MLOps pipeline to proceed with standard isolated retraining. Moderate Risk: The agent directs the pipeline to proceed with upstream retraining but concurrently triggers intensified monitoring of downstream components (using integrated tools such as Prometheus and Grafana [2]) and may prepare contingency plans. High Risk: The agent activates more sophisticated, defensive strategies:

- Option 1 (Agent-flagged Expert Review): The agent pauses immediate retraining and flags the situation for human expert diagnosis. Furthermore, a crucial aspect will be to establish a systematic feedback loop where insights derived from these human expert reviews are formally captured. This captured knowledge will then be used to iteratively refine the underlying rules or update the training data for both the “Entanglement Forecaster” and the agent’s policy logic, fostering continuous improvement.
- Option 2 (Agent-orchestrated Coordinated Retraining): The agent initiates a “team retraining,” orchestrating the update of both the upstream model and its directly affected downstream neighbors. The agent system would manage the complex dependencies and sequencing for this operation. While the operational complexity scales with the number of involved components, future extensions could explore strategies such as graph-based pruning to prioritize only the most critically impacted downstream components for coordination, or investigate batched/staged coordinated retraining rollouts to manage resource demands for very large systems.
- Option 3 (Agent-selected Constrained Retraining): The agent directs the MLOps pipeline to apply gentler, constrained retraining techniques to the upstream model.

Agent Logic Complexity & Feasibility: The initial implementation of the agent’s decision logic can be rule-based (low computational complexity, $O(1)$ post-risk assessment), making it highly feasible. For more advanced adaptability, future work could explore learning-based agents (using, for instance, Reinforcement Learning frameworks like RLlib, integrated with Python). This would increase implementation complexity but could provide more nuanced decision-making over time. The primary operational complexity stems from the actions the agent triggers, particularly Coordinated Retraining, which scales with the number of involved components.

Phase 3: The Moment of Truth - Rigorous Evaluation

This phase scientifically demonstrates whether this agent-driven, adaptive framework is genuinely more effective than fixed rules, especially in complex AI systems.

- **Setting the Stage (Experimental Setup):** The MLS simulator will be used for comprehensive experiments, varying parameters such as dataset drift and entanglement probabilities (inspired by concepts from [4]).
- **The Competition (Baseline Comparison):** The Entanglement-Aware MLOps framework (with its agent-based orchestrator) will be compared against: A Standard Progressive Policy (retraining individual components on local failure [4]). A Standard Conservative Policy (retraining on overall system failure [4]).
- **The Scorecard (Key Evaluation Metrics):** System Reliability (Service Availability): Proportion of time the MLS operates correctly. [4] A statistically significant reduction in performance

problems will be prospectively defined, aiming, for instance, for a quantifiable decrease (such as targeting a 20-30% lower rate of entanglement-induced system failures) or a measurable improvement in key operational metrics like system uptime when benchmarked against baseline performance. Entanglement Incidents: Frequency and severity of upstream “fixes” causing downstream problems. Smart Resource Use (Resource Efficiency): Reduction in computational power used for ineffective retraining. [1] Adaptive Behavior Deep-Dive: Analysis of how the “Smart Navigator” agent makes choices.

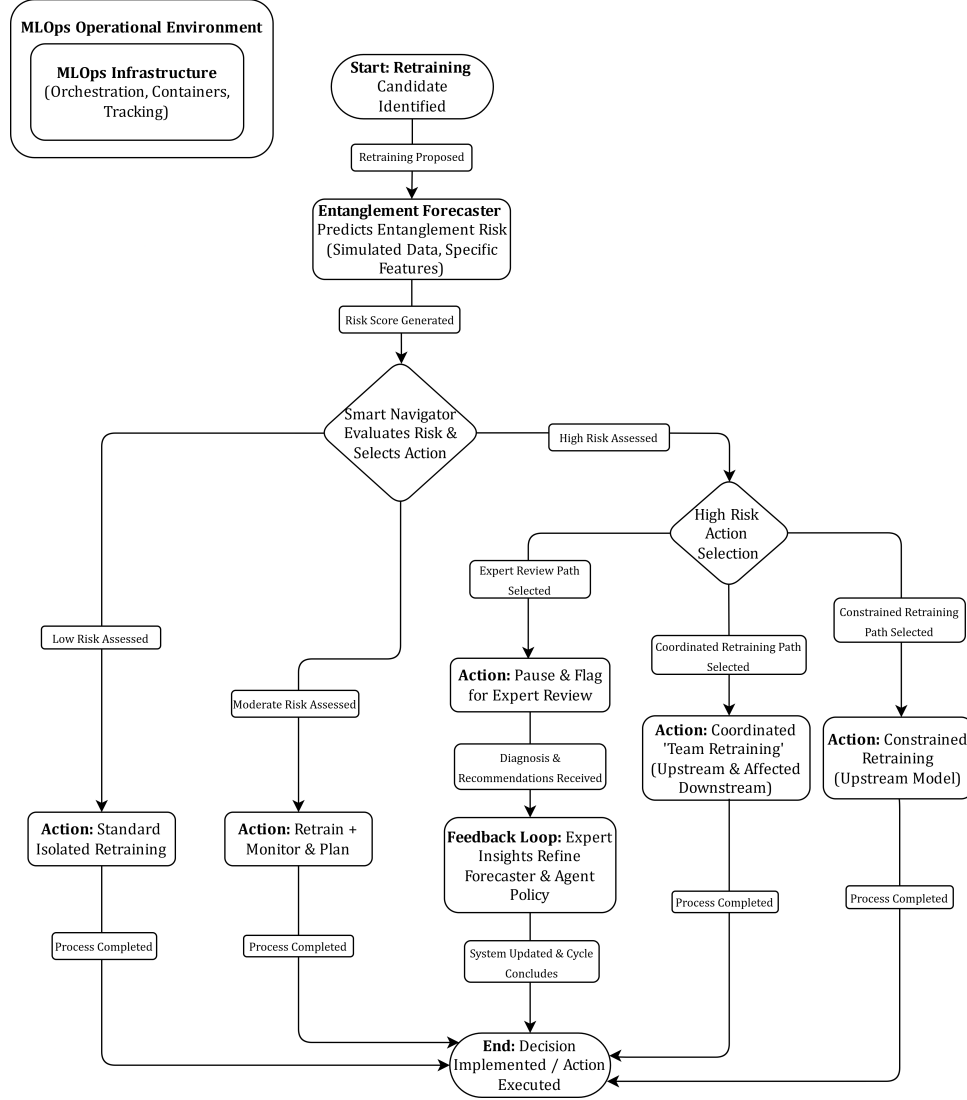


Figure 1: Proposed Workflow Diagram

Overall Feasibility of the Proposal:

This research, while aiming for an intelligent MLOps system, is designed for feasibility by:

1. Starting with rule-based agents and potentially exploring learning agents iteratively.
2. Leveraging mature open-source technologies for MLOps, simulation, and ML. [2]
3. Using a simulation-first approach for development and rigorous testing.
4. Maintaining a sharp focus on mitigating entangled enhancement through predictive and adaptive means.

This approach aims to provide strong evidence for an MLOps system that actively anticipates and adapts to entangled enhancement, extending beyond static policy analyses. [4]

Expected Contributions

Anticipated outcomes include:

1. A novel predictive model (“Entanglement Forecaster”) for estimating the risk of entangled enhancement prior to upstream model retraining in multi-component MLS.
2. A novel adaptive retraining policy, executed by an intelligent agent, that integrates entanglement risk predictions into MLOps workflows to enhance decision-making flexibility and system reliability.

Conclusion

This research proposes a novel, predictive, and adaptive MLOps framework to proactively address entangled enhancement in complex Machine Learning Systems. By integrating a risk-forecasting model with an intelligent retraining policy, the system aims to mitigate self-defeating updates, enhance overall stability, and improve resource efficiency. The simulation-driven methodology ensures feasibility, scalability, and rigorous evaluation against traditional static policies. The outcome is expected to advance the intelligence and resilience of modern AI operations.

References

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